Md. Jaied Hasan

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Academic Credentials

Bangladesh University of Engineering and Technology (BUET), Dhaka Jan 2022 – Jun 2026
Bachelor of Science in Electrical and Electronic Engineering **CGPA: 3.80/4.00** (Top 27%)
Major: Communication and Signal Processing *Last four semesters: 3.85/4.00*

Research Experience

Bangladesh University of Engineering and Technology (BUET), EEE Oct 2026 – Present
Research Assistant | Supervisor: Dr. Md. Forkan Uddin, Professor, EEE, BUET

- Conducting research on Wi-Fi 8 (IEEE 802.11bn) multi-AP coordination, with emphasis on coordinated beamforming (Co-BF), multi-AP optimization, and performance under inter-cluster interference (ICI).
- Developed a Python-based multi-AP simulator to compare three operating modes: conventional CSMA/CA with beamforming, Co-BF with ICI, and Co-BF without ICI.
- Investigating AP recruitment/pairing and multi-AP coordination strategies, with particular focus on how ICI changes the throughput and interference behavior of coordinated deployments.

Undergraduate Thesis: Federated Sparse PCA with Differential Privacy Using Correlated Noise
Supervisor: Dr. Hafiz Imtiaz, Professor, EEE, BUET

- Developed a federated sparse-PCA framework for horizontally partitioned datasets.
- Investigated correlated-noise mechanisms for differentially private dimensionality reduction and feature selection.
- Evaluated privacy–utility, convergence, sparsity, reconstruction, and communication-efficiency trade-offs across distributed settings.
- Implemented reproducible experimental pipelines in Python using NumPy, Pandas, and scientific-computing tools.
- Applied linear algebra, optimization, statistical analysis, and algorithmic evaluation to privacy-preserving distributed learning.

Publications

- **Journal manuscript under review (first author):** "Federated Sparse PCA with Differential Privacy Using a Correlated-Noise Mechanism." *The Journal of the Franklin Institute* (Q1).
- **Conference paper accepted (2026; first author):** "An Integrated Dual-Module Cyber Physical System for Motorcycle Safety: Utilizing Pre-Ride Safety, Crash Detection and Swift Response with Mobile Application Companion." *IEEE ICACCESS 2026* (Acceptance rate 23.3%).
- **Conference submission (2026):** "ML-Based Mobile Traffic Forecasting for Network-Capacity Planning." Submitted to *IEEE ICECE 2026*.

Technical Skills

Programming	Python, C, C++, MATLAB, Verilog
ML / Deep Learning	PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, XGBoost, Random Forest, Logistic Regression, XLM-RoBERTa, MobileNetV2, YOLOv8, KNN
Trustworthy / Efficient AI	Federated learning, differential privacy, sparse PCA, knowledge distillation, class-weighted learning, label smoothing, model calibration, mixed-precision training, group-aware validation, data-leakage auditing
NLP	Multilingual/code-mixed NLP, byte-level BPE tokenization, TF-IDF (word/character n-grams), embeddings, teacher–student learning
Computer Vision	OpenCV, face recognition, object detection
Signal / Data Processing	Digital signal processing, MFCC, time-series analysis/forecasting, PCA/SPCA, NumPy, Pandas
Wireless / Networking	Wi-Fi 8 (IEEE 802.11bn), multi-AP coordination, coordinated beamforming (Co-BF), MRT/ZF/regularized-ZF precoding, inter-cluster interference analysis, discrete-event CSMA/CA and NAV modeling, wireless-network optimization
Embedded / IoT / Edge	ESP32, ESP8266, Arduino, Raspberry Pi, ESP32-CAM, Pixhawk (PX4), IMU/MPU6050, GPS, GSM/SIM800L, ESP-NOW, embedded web servers, finite-state-machine firmware
Protocols	I2C, SPI, UART
Software / Engineering Tools	Linux, Git/GitHub, ROS, MATLAB/Simulink, Proteus, PSpice, PSAF, AutoCAD, KiCad, MIT App Inventor

Projects

Machine Learning, Data Science, and Software

Multilingual Toxicity Classification with Knowledge Distillation
HerWILL AI for Digital Safety Datathon 2026 · XLM-RoBERTa · Hugging Face Transformers · Knowledge distillation · Byte-level BPE

- Developed an XLM-RoBERTa teacher–student pipeline for Bangla, English, and Banglish/code-mixed harmful-content classification.
- Fine-tuned the teacher on a 47K+ sample, three-class dataset using stratified 5-fold out-of-fold training, with class-weighted training, label smoothing, mixed precision, temperature scaling, and macro-F1 bias calibration.
- Designed a compact ∼5M-parameter student with a custom byte-level BPE tokenizer and mean/max pooling, distilled from temperature-scaled teacher probabilities.
- Achieved ∼55× parameter compression relative to the teacher; the distilled student pipeline scored 0.344 F1NOP in the datathon.

ML-Based Mobile Traffic Forecasting and Dynamic Bandwidth Planning Code
Python · XGBoost · LightGBM · TensorFlow/Keras (LSTM, GRU, RNN, 1D-CNN) · statsmodels

- Developed a forecasting pipeline for regional mobile-internet traffic using BTCL data, with emphasis on the Savar region, working with hourly link-throughput (Gbps) series for two operators (Grameenphone and Robi).
- Engineered calendar, lag (1, 2, 3, 24 h), and 24-h rolling-statistic features plus exogenous event flags (weekends, holidays, operator offers, public events, political gatherings, power cuts, rain); examined trend and seasonality with STL decomposition, FFT, and ADF/KPSS stationarity tests.
- Benchmarked XGBoost, LightGBM, Random Forest, LSTM, GRU, SimpleRNN, 1D-CNN, AR, ARIMA, and Holt-Winters models on chronological splits with time-series cross-validated tuning, using MAE and RMSE, including feature-free, autoregressive, and cross-operator experiments.
- A paper based on this work was submitted to IEEE ICECE 2026.

Leakage-Aware Content-Refresh Prioritization – FlyRank ML Internship Code
Python · Scikit-learn · Pandas · Jupyter/Colab

- Framed content-refresh triage as a ranking problem evaluated by Precision@K, and built a future-decline target with separated feature (Mar 1–15), decision (Mar 16), and outcome (Mar 17–31) windows over 58,097 eligible pages.

- Compared a hand-written rule baseline, Logistic Regression, and Random Forest on a client-grouped holdout (26 training vs. 7 unseen test clients, zero overlap); Logistic Regression raised Precision@100 from 0.70 to 0.87 against a 36.1% base rate.
- Audited validation and leakage by contrasting page-random and client-grouped splits, confirming that no post-decision features were used, and running a label-sensitivity ablation (Precision@100 fell to 0.37 without impression-derived predictors); rewrote result claims to match this evidence.

Lung Disease Classification from Respiratory Audio Code
MATLAB · MFCC · K-Nearest Neighbors

- Built a MATLAB pipeline that classifies lung-sound recordings as normal, asthma, or heart failure, with resampling, normalization, and fixed 10-s segment extraction before feature computation.
- Extracted Mel-Frequency Cepstral Coefficient (MFCC) features and trained a K-Nearest Neighbors classifier on an 80/20 split of two merged open-source respiratory-sound datasets, achieving 98.66% accuracy (98.66% macro sensitivity).

Face-Recognition Attendance System Code
Python · OpenCV · face_recognition · Tkinter

- Built a webcam-based attendance system with a Tkinter GUI for face enrollment and real-time recognition using face encodings and nearest-match face-distance identification.
- Logged timestamped attendance to CSV, recording each recognized person once per session.

Text-Classification Inference API (Text Evaluator) Code
Python · Scikit-learn · Flask

- Trained a text classifier that combines character-level and word-level TF-IDF n-gram features with a class-balanced Logistic Regression model.
- Deployed the model as a CORS-enabled Flask REST service with `/predict` (class label and probabilities) and `/health` endpoints for browser front-ends.

Personal Academic Portfolio Website Code
JavaScript · HTML/CSS · GitHub Pages

- Built and maintain a single-page academic portfolio (live site) with searchable, filterable project and publication sections and a multi-photo lightbox, with all content driven from a single data file.

Wireless Communications and Signal Processing

Wi-Fi 8 Multi-AP Coordinated Beamforming Simulator Code
Python · NumPy · Discrete-event simulation · Monte Carlo analysis

- Built an explainable, geometry-based, discrete-event downlink simulator for IEEE 802.11bn multi-AP coordination, modeling CSMA/CA, AP pairing, physical carrier sensing, omnidirectional NAV advertisement, and per-AP throughput accounting; complements ongoing BUET research.
- Implemented 4-element ULA beamforming with MRT, zero-forcing, and MMSE-style regularized-ZF precoders under per-AP conducted-power constraints, over line-of-sight path-loss channels with SINR computed from all concurrent streams.
- Compared three modes (CSMA/CA with MRT, two-AP Co-BF with inter-cluster interference, and an ICI-free counterfactual) and produced beam-pattern, EIRP, SINR-heatmap, and ICI-penalty diagnostics, Monte Carlo and precoder sweeps, and smoke tests that validate the PHY model.

Frequency Division Multiplexing (FDM) Hardware Demonstration
Analog communication · Discrete components

- Constructed an analog FDM hardware demonstration from discrete components, transmitting multiple signals simultaneously over a shared medium.

Embedded Systems, IoT, and Robotics

HIVE-R: Bio-Inspired GPS-Free Rescue Drone for Disaster Response Code

Raspberry Pi 4 · Pixhawk (PX4) · MAVLink · YOLOv8n · Visual odometry/SLAM

- Developed a GPS-independent, semi-autonomous search-and-rescue drone system for post-disaster environments, with an architecture inspired by honeybee swarms in which "worker" drones search sub-areas and a ground-control station (GCS) aggregates victim reports.
- Implemented camera–IMU (webcam and MPU6050) visual odometry/SLAM on a Raspberry Pi 4 companion computer linked to a Pixhawk (PX4) flight controller over MAVLink for GPS-denied localization.
- Deployed a YOLOv8 Nano human detector trained on a CMU human-detection dataset with degradation augmentation, reporting 93.2% precision, 91.5% recall, and 92.1% mAP@0.5 at ∼180 ms per frame on the Raspberry Pi 4; victim coordinates are relayed to the GCS over 915 MHz radio telemetry.

LifeSaver Wearable Smart Helmet Code

ESP32 · ESP8266 · ESP-NOW · SIM800L · NEO-6M GPS · MPU6050 · MIT App Inventor

- Developed a two-module helmet/bike safety system: an ESP32 helmet unit and an ESP8266 bike unit communicating over ESP-NOW, with a relay interlock that blocks ignition unless the helmet is worn (IR sensor) and no alcohol is detected (MQ-3).
- Implemented threshold-based fall detection from MPU6050 acceleration and angular-velocity signals, automatic and manual SOS SMS alerts with GPS coordinates over SIM800L GSM, and a GPS-based overspeed alarm above 80 km/h.
- Built a Bluetooth Android companion app for emergency-number configuration, live system status, phone-side SOS, and GSM fallback; this work underlies the accepted IEEE ICACCESS 2026 paper.

Autonomous Fire-Extinguishing Robot Code

Arduino UNO · ESP32 · ESP8266 · ESP-NOW · L298N · C++

- Built an autonomous mobile robot that locates fire direction with three flame sensors, navigates with L298N-driven DC motors, and suppresses flames with a servo-aimed water-spray nozzle in a two-room test environment.
- Implemented closed-loop flame-directed motion, a safety retreat when temperature exceeds 140°C, and ESP-NOW telemetry between ESP32 and ESP8266 controllers (Control Systems Lab, BUET).

ESP-Eye: Real-Time Object Recognition on an ESP32-CAM Stream Code

Python · OpenCV · TensorFlow/Keras · MobileNetV2 · ESP32-CAM

- Tested whether a low-cost ESP32-CAM video stream can support deep-learning inference: captured the Wi-Fi stream with OpenCV and ran an ImageNet-pretrained MobileNetV2 on every fifth frame, overlaying the top prediction and confidence score in real time.

Dual-Channel Pressure Monitoring System Code

ESP32 · C++ · I2C LCD · HTML/JavaScript dashboard

- Built an ESP32 device that monitors two analog pressure sensors (0–200 kPa) with kPa/mmHg display units, per-channel thresholds saved in non-volatile memory, LED/buzzer alarms, and a 16×2 LCD with a three-button, two-mode interface.
- Developed a Wi-Fi web dashboard with 1 Hz live readings, remote threshold configuration, and an NTP-timestamped event log with CSV export.

ESP32 Smart Voltage and Overcurrent Protection System Code

ESP32 · C++ · Finite-state machine · Embedded web server

- Implemented protection logic as a finite-state machine (INIT, NORMAL, TRIPPED, RESET_READY) with latched faults and a 1-s recovery check before manual reset; default limits 250 V and 8 A on simulated voltage/current inputs.
- Provided local control via a 16×2 I2C LCD and pushbuttons, plus an ESP32 access-point web dashboard for live readings, fault reasons, adjustable limits, and remote reset; documented with a design report, FSM diagram, schematics, and test evidence.

Electronics, Power Systems, and Electrical Design

PCB Design Collection Code

KiCad · Schematic capture · PCB layout · Gerber fabrication files

- Designed and published KiCad schematics, board layouts, and Gerber/drill fabrication files for an AC-to-DC full-bridge rectifier with smoothing and an indicator LED, a compact 70×40 mm transformerless 110–220 V AC to 5 V DC supply, and a potentiometer-controlled PWM servo-motor tester.

Digital-Logic Tic-Tac-Toe Code

Digital logic ICs · D flip-flops · Proteus

- Designed and built a microcontroller-free Tic-Tac-Toe game from 36 logic ICs (D flip-flops; AND, NOR, NOT, and XOR gates; counters), with push-button moves and reset, LED game-state indicators, and two 7-segment scoreboards.

Frequency Converter with Real-Time Display and Tuning

AC–DC–AC conversion · Signal conditioning · LCD

- Designed a low-cost AC signal conditioner that converts arbitrary AC waveforms into a stable sinusoidal output with tunable frequency, using an AC–DC–AC architecture with live frequency readout on an LCD.

Transient Analysis App

MATLAB

- Built a MATLAB application that simulates AC and DC circuits, plots time- and frequency-domain responses, and visualizes steady-state phase differences.

Sensitivity Analysis of IEEE 30- and 33-Bus Systems

PSAF · MATLAB

- Simulated cascading failures in IEEE 30-bus and 33-bus power networks to identify critical nodes and study system sensitivity.

Linear Induction Motor (LIM)

Electromechanical design · Winding · Characterization

- Designed, wound, and characterized both two-pole and four-pole linear induction motor configurations.

Electrical Service Design for a 14-Story Residential Building

AutoCAD · Load scheduling · Distribution planning

- Produced a complete electrical service layout for a 14-story, four-unit residential apartment building, including load schedules, wiring diagrams, and distribution planning.

Research and Technical Interests

- Trustworthy and Privacy-Preserving ML
- Federated and Distributed Learning
- Efficient and Scalable AI
- Robust Machine Learning
- Multimodal and Foundation Models
- AI for Healthcare
- Deep Learning
- Signal Acquisition and Processing
- Wi-Fi 8 (IEEE 802.11bn) and Multi-AP Coordination
- Coordinated Beamforming (Co-BF) and Interference-Aware WLAN Optimization
- Wireless Communications and Network Optimization
- Directional Channel and Beamforming Analysis
- Wireless Networks, IoT, and Edge Intelligence
- Data-Driven Communication Systems
- Cyber-Physical Systems
- Robotics and Automation
- Mechatronics and Embedded Systems
- Human–Robot Interaction
- Embedded Systems and Edge Devices

Relevant Coursework

- Artificial Intelligence and Machine Learning
- Machine Learning Specialization (Coursera)
- Digital Signal Processing
- Introduction to Medical Imaging
- Probability and Statistics
- Wireless Communications
- Communication Systems I and II
- Microprocessor and Embedded Systems
- Robotics and Automation
- Control Systems
- Electronic Circuits I and II
- Digital Electronics
- DC/AC Circuits
- Power Electronics
- Structured Programming in C/C++

Professional Experience

FlyRank AI Aug 1, 2026 – Present
Machine Learning Intern *Remote*

- Developed an ML pipeline to detect and rank declining web content across 23,000+ webpages using historical traffic and trend-based features.
- Performed feature engineering, Precision@K evaluation, error analysis, and model diagnostics to translate predictions into actionable content-review recommendations.
- Strengthened model reliability through time/group-aware validation, data-leakage audits, and failure analysis, aligning reported conclusions with experimentally supported evidence.

Banglalink Digital Ltd. 18 Nov 2025 – 25 Nov 2025
Industrial Attachment

Bangladesh Mathematical Olympiad Nov 2022 – Dec 2025
Academic Team Member

Honors and Awards

- **Dean's List Scholarship** – awarded for excellent academic performance in Level 1 and Level 3.
- **University Merit Scholarship** – awarded for excellent academic performance.
- **Shamsul Huda Chowdhury Scholarship** – BUET trust-fund scholarship for students with outstanding academic records.
- **Professor Dr. Ahmad Ullah and Mrs. Chemanara Ahmad Ullah Trust Fund Scholarship** – awarded for outstanding academic performance at BUET.
- **1st Runner-Up, Brain of Varendra** – talent-hunt competition organized by Varendra University, Rajshahi.